

Correlation

Reading Objectives:

In this reading, you will learn about the methods to find the relation between two variables and the strength of the relation. You will also learn about covariance and coefficient of correlation to find the relation.

Main Reading Section:

1. Covariance:

The statistical technique used for studying the existence and extent of relationships among two or more variables is the covariance, which is given by

$$\text{Covariance}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

So, in bivariate and multivariate cases, the statistical technique used for studying the existence of a relationship between all variables is covariance. Covariance signifies the direction of the linear relationship between the two variables.

By direction, we mean if the *variables* are directly proportional or inversely proportional to each other. (Increasing the value of one variable might have a positive or a negative effect on the value of the other variable.)

The values of covariance can be any number between the two opposite infinities. Also, it is important to mention that covariance only measures how two variables change together, not the dependency of one variable on another one.

Even though the covariance explains the relationship between two or more variables, the units of the measurement of the variable are attached to the value of the covariance. If they have different units of measurement, based on the sign, it is possible to say whether they are related or not, but the degree (strength) of the relationship is not possible to decide based on the value of covariance due to different units of measurement.

Example: Find the covariance between age (months) and weight (kgs) for the following data:

Age (years)	7	6	8	5	6	9
Weight (kgs)	12	8	12	10	11	13

Solution:

Age(X)	Weight(Y)	$X - \bar{X}$	$(y - \bar{y})$	$(X - \bar{X})(y - \bar{y})$
5	20	-2	-3	6
4	18	-3	-5	15
7	23	0	0	0
10	32	3	9	27
8	26	1	3	3
8	25	1	2	2
$\sum X = 42$	$\sum Y = 144$			$\sum (X - \bar{X})(y - \bar{y}) = 53$

$$\text{Cov}(X, Y) = 53/5 = 10.6 \text{ years/kg}$$

Interpretation: Age and weight are directionally proportional.

If X and Y are two random variables with probability mass function p(x)/probability density function f(x), then the covariance between X and Y is given by

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - E(X)E(Y), \text{ where } E(X) = \mu_X \text{ and } E(Y) = \mu_Y \text{ are the means of X and Y.}$$

2. Coefficient of Correlation:

The correlation coefficient is a statistic that assesses the strength and direction of the relationship between two continuous variables, which is given by

$$r = \frac{\text{Covariance (x, y)}}{\sqrt{\text{Var (x)}} \sqrt{\text{Var(Y)}}}$$

$$\frac{\sum_{i=1}^n (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum_{i=1}^n (X - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y - \bar{Y})^2}}$$

If the sign is positive, it means the relation is direct (an increase in one variable is associated with an increase in the other variable, and a decrease in one variable is associated with a decrease in the other variable).

If the sign is negative, it means an inverse or indirect relationship (an increase in one variable is associated with a decrease in the other).

The value of r ranges between (-1) and (+1), i.e., $-1 \leq r \leq +1$

Interpretation based on r value.

r = - 1: Perfect negative correlation

r = - 0.99 to - 0.76: Strong negative correlation

r = - 0.75 to - 0.26: Intermediate (moderate) negative correlation

r = - 0.25 to 0: Weak negative correlation

r = 0: Zero correlation

r = 0 to 0.25: Weak positive correlation

r = 0.26 to 0.75: Intermediate (moderate) positive correlation

r = 0.76 to 0.99: Strong positive correlation

r = 1: Perfect positive correlation

Example: Find the correlation coefficient of the following and interpret the relation.

Age (years)	7	6	8	5	6	9
Weight (kgs)	12	8	12	10	11	13

Solution:

$$\frac{\sum_{i=1}^n (X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum_{i=1}^n (X - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y - \bar{Y})^2}}$$

r = 0.759

Positive (direct) strong correlation